

ORIGINAL RESEARCH

Soluble Thrombomodulin Links Viremia and Mortality During COVID-19: Results From the ACTIV-4a Trial

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BACKGROUND: COVID-19 and other respiratory viral infections can cause cardiovascular complications. SARS-CoV-2 virions are found in the blood, and circulating viral RNA levels are associated with death. We hypothesized that viremia can induce thrombotic endotheliopathy that contributes to death and studied this relationship in patients hospitalized for COVID-19 and enrolled in the ACTIV-4a (Accelerating COVID-19 Therapeutic Interventions and Vaccines) randomized trial of antithrombotic therapy.

METHODS: We quantified SARS-CoV-2 nucleocapsid RNA and protein in plasma and measured their associations with clinical outcomes and biomarkers of thromboinflammation and endotheliopathy. We used Cox regression and Fine-Gray competing risk models to analyze survival and thrombosis. We conducted causal mediation analysis to explore whether thrombotic endotheliopathy mediates the relationship between viral RNA and death.

RESULTS: In 93 patients, SARS-CoV-2 RNA and N-antigen were higher in nonsurvivors. Baseline viral RNA levels were associated with increased 90-day death (hazard ratio [HR], 1.31 [95% CI, 1.17–1.46]) and thrombosis (HR, 1.35 [95% CI, 1.24–1.47]). Viral RNA more effectively predicted survivorship than N-antigen levels. Soluble thrombomodulin, a biomarker of thrombotic endotheliopathy, positively correlated with viral RNA levels. Mediation analysis revealed that soluble thrombomodulin accounts for 12.2% (95% CI, 0.1%–32.3%; $P=0.048$) after adjustment for age and sex of the relationship between viral RNA and the 90-day mortality rate.

CONCLUSIONS: Elevated plasma SARS-CoV-2 RNA levels are associated with death in ACTIV-4a, which is causally mediated in part by soluble thrombomodulin. We propose that lung–blood viral dissemination is a potential mechanism for cardiovascular complications of respiratory viruses.

REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique Identifier: NCT04505774.

Key Words: endotheliopathy ■ mediation analysis ■ SARS-CoV-2 ■ thrombosis ■ viremia

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RESEARCH PERSPECTIVE

What Is New?

- Circulating viral RNA, or viremia, during COVID-19 infection requiring hospitalization is associated with increased thrombotic events and death during the ACTIV-4a (Accelerating COVID-19 Therapeutic Interventions and Vaccines) clinical trial.
- Viral RNA levels are also associated with soluble thrombomodulin, which mediates a portion of the relationship between viral RNA and death, suggesting that viremia may induce endotheliopathy, leading to worse clinical outcomes.

What Question Should Be Addressed Next?

- Future research may consider whether viremia and soluble thrombomodulin during respiratory viral infections other than COVID-19 are associated with worse outcomes and, if so, whether they may be therapeutically modifiable to improve cardiovascular outcomes during and after respiratory viral infection.

Nonstandard Abbreviations and Acronyms

ACTIV-4a	Accelerating COVID-19 Therapeutic Interventions and Vaccines
ACTT-1	Adaptive COVID-19 Treatment Trial
ATTACC	Antithrombotic Therapy to Ameliorate Complications of COVID-19
REMAP-CAP	A Randomized, Embedded, Multifactorial, Adaptive Platform Trial for Community-Acquired Pneumonia
sRAGE	soluble receptor for advanced glycation end products
sTM	soluble thrombomodulin
THBD	thrombomodulin gene

Respiratory viral infections such as influenza and COVID-19 can cause cardiovascular complications that may result from increased thrombosis¹⁻³ and endothelial injury.⁴ However, the mechanism(s) by which viral infection in the lower respiratory tract can lead to cardiovascular complications is uncertain. We and others have demonstrated that SARS-CoV-2 virions can be found in the blood during infection,⁵ and

the level of circulating viral RNA correlates with clinical outcomes, including disease severity and death.⁵⁻¹³ We have also shown that plasma RNA levels positively correlate with lower respiratory tract viral RNA levels.¹⁴ Therefore, the translocation of respiratory tract virions from the lung into blood offers a potential pathway to understand the cardiovascular and thrombotic manifestations of viral infection.

Cardiovascular complications are associated with increased death in patients with COVID-19.^{2,15} The endothelium is a crucial component of the interface between the airspaces and blood, and studies have shown the presence of endothelial injury and in situ immunothrombosis in pulmonary and cardiovascular tissue samples from deceased patients with COVID-19.^{4,16-19} There may also be direct endothelial injury during COVID-19,²⁰ as others have identified SARS-CoV-2 virions in endothelium with associated microangiopathy and in atherogenic plaques with associated inflammation.¹⁸ Therefore, viral-mediated endothelial injury and vascular inflammation may contribute to the risk of cardiovascular complications and death in patients with COVID-19.

We used plasma samples collected from patients enrolled in the randomized ACTIV-4a (Accelerating COVID-19 Therapeutic Interventions and Vaccines) trial of therapeutic anticoagulation during inpatient hospitalization for COVID-19.^{21,22} We quantified plasma levels of SARS-CoV-2 RNA and protein and measured their association with clinical outcomes and key biomarkers to test the hypothesis that endothelial injury mediates the known association between plasma viremia and death during COVID-19.

METHODS

This study is a secondary analysis of deidentified data and biospecimens from the ACTIV-4a trial (ClinicalTrials.gov: NCT04505774), which was approved by Western Institutional Review Board (Registration No. IRB00000533). All participants provided written informed consent as part of the parent trial. The data and analysis that support the findings of this study are available from the corresponding author upon reasonable request.

Study Cohort

The ACTIV-4 Mechanistic Studies Network through the Collaborating Network of Networks for Evaluating COVID-19 and Therapeutic Strategies program of the National Heart, Lung, and Blood Institute provided plasma biospecimens and deidentified clinical data from a subset (N=93) of patients hospitalized with COVID-19 enrolled in the ACTIV-4a clinical trial. Eligibility criteria for ACTIV-4a trial participants are provided in [Tables S1](#) through [S4](#), and the subgroup

described herein was similar to the overall ACTIV-4a cohort (Table S2). This cohort included patients with both severe and nonsevere disease and included all treatment arms including anticoagulation, antiplatelet, and control arms.

SARS-CoV-2 Quantification

Plasma samples were collected according to ACTIV-4a protocol on enrollment day (day 0, baseline) and on days 3, 7, and 14 if patients remained hospitalized and alive. We performed quantitative real-time polymerase chain reaction for SARS-CoV-2 as previously described for plasma samples.⁵ Briefly, we extracted total RNA from 0.25 mL of EDTA plasma using the MagMax Viral Pathogens Kit (Thermo Fisher Scientific, Waltham, MA), and performed 1-step quantitative real-time polymerase chain reaction of the SARS-CoV-2 N gene and human *RNaseP* gene, as previously described, using the following N-specific primers (forward: 5'-GTTTGGTG GACCCTCAGATT-3', reverse: 5'-CGCAGTATTATTGGGT AAACCTTG-3', Probe: 5'-6-FAM-TAACCAGAATGGAGAAC GCAGTGGG-3'BHQ1). We also quantified SARS-CoV-2 N-antigen in EDTA plasma samples using a commercially available assay according to the manufacturer's instructions (R-PLEX SARS-CoV-2 N Assay; Meso Scale Discovery, Rockville, MD).²³

Biomarker Analysis

We quantified coagulation proteins including thrombin-antithrombin complexes (Affinity Biologicals, Ancaster, ON, Canada; No. TAT-EIA), fibrinogen (Affinity Biologicals; No. FG-EIA), and soluble thrombomodulin (sTM; R&D Systems, Minneapolis, MN; No. DTHBD0) in citrated plasma samples by sandwich ELISA according to manufacturer instructions as previously described.²⁴ tPA (tissue plasminogen activator) antigen was also quantified as previously described.²⁵ We quantified 9 lung injury host-response biomarkers in EDTA plasma using a customized Luminex assay (R&D Systems).^{26–29}

Clinical Data Extraction and Definitions

We analyzed baseline demographics along with key clinical time points including date of ACTIV-4a trial enrollment, date of first and subsequent thrombotic events, clinically measured concentrations of D-dimer, and date of death. To account for differences in clinical D-dimer assays used at different trial sites, D-dimer values measured in fibrinogen equivalent units were multiplied by 2 to adjust to D-dimer units, which are reported herein. We derived the World Health Organization 10-point ordinal scale of COVID-19 severity, time-to-clot, duration of hospitalization, and 90-day survival. Classification of disease severity in the ACTIV-4a trial

was based on the provision of intensive care unit-level organ support rather than by strict intensive care unit admission criteria.²¹

Statistical Analysis

We compared plasma RNA and N-antigen levels across time points and clinical groups, including 90-day survival and thrombosis status, with nonparametric Wilcoxon tests. We calculated Spearman correlation coefficients for plasma markers of SARS-CoV-2 viremia and biomarkers. We corrected for multiple comparisons using the Benjamini-Hochberg method. Linear mixed-effects models with random intercepts for patients were used to compare log-transformed viremia biomarkers between clinical groups across study days. Models included fixed effects for death status and study day (categorical) and their interaction and were estimated using restricted maximum likelihood. The model can be written as:

$$y_{ij} = \beta_0 + \beta_1 \text{death}_{ij} + \beta_2 \text{StudyDay}_{ij} + \beta_3 (\text{death}_{ij} \times \text{StudyDay}_{ij}) + b_{0i} + \varepsilon_{ij},$$

where i indexes patients, j indexes observations, b_{0i} is a subject-specific random intercept, and ε_{ij} is the residual error.

We performed unadjusted and multivariable-adjusted Cox regression analysis for death as well as Fine-Gray competing risk regression analysis for thrombosis with competing risk of death across disease severities.³⁰ Continuous biomarkers were standardized via Z score scaling before Cox and Fine-Gray modeling.

Single-Nucleus Sequencing Analysis

We analyzed single-nucleus RNA sequencing results from a COVID-19 tissue atlas (Gene Expression Omnibus; GSE171668). We used CellBender count matrices and linked count matrices to relevant meta-data that included automated predicted cell-type annotations. If cell type was missing, it was recorded as undefined. All analyses were done in R (R Foundation for Statistical Computing, Vienna, Austria) using Seurat. Counts within each organ object were normalized using Seurat log normalization with a scale factor of 10 000. The highly variable features were identified using FindVariableFeatures and then scaled with ScaleData. The GRCh38.premrna thrombomodulin gene (*THBD*) expression values were extracted and grouped by predicted cell type to form violin plots. If cells had raw counts >0, they were considered *THBD* positive. SARS-CoV-2 RNA transcripts in *THBD*-expressing and nonexpressing cells were compared using Fisher's exact test.

Causal Mediation Analysis

We performed causal mediation analyses using the “mediation” package in R.³¹ The mediator was modeled using linear regression, and the outcome was modeled using logistic regression or penalized logistic regression when low event rates led to model nonconvergence. Models were fit with and without adjustment for age and sex. The average causal mediated effect, average direct effect, and total effect were estimated using nonparametric bootstrap resampling (500–1000 simulations) to obtain CIs. Models were structured as follows:

$$M = \alpha_0 + \alpha_1 X + C\alpha$$

$$\text{logit}\{P(Y = 1)\} = \beta_0 + \beta_1 X + \beta_2 M + C\beta$$

where M=mediator (sTM), X=SARS-CoV-2 viral load, Y=death, and C=covariates (age and sex).

RESULTS

Demographics

Ninety-three patients enrolled in the ACTIV-4a trial between November 9, 2020, and September 21, 2021, with 264 total patient time points were included in our analysis (Table 1). We were provided baseline samples from 88 patients. Forty-five (48%) patients with baseline data met trial criteria for “severe” disease on the basis of the provision of intensive care unit-level care. Forty-three patients were classified as “nonsevere” at baseline. In the severe group at baseline, 38 (84%) subjects met criteria for World Health Organization ordinal class 6, 1 (2%) for class 7, 3 (7%) for class 8, and 3 (7%) for class 9. Notably, there are differences in therapeutic response to antithrombotic therapy by disease severity during COVID-19.^{21,22} In our cohort, patients in the severe group experienced longer hospital stays (median, 14 [interquartile range, 9.0–24.0] versus 5.0 [interquartile range, 4.0–9.0] days; $P<0.001$), increased 90-day mortality rate (14 [31.1%] versus 4 [9.3%]; $P=0.016$), and increased frequency of thrombotic events that did not reach statistical significance (10 [22.2%] versus 3 [7.0%]; $P=0.07$). There were no intergroup differences in rates of underlying cardiovascular comorbidities, diabetes, or immunosuppression.

Plasma SARS-CoV-2 RNA Levels Are Associated With Death

We quantified SARS-CoV-2 nucleocapsid RNA and N-antigen protein levels in available specimens. RNA was detectable in 103 (39%) patient time points with a median level of 263.5 copies/mL (interquartile range, 54–1556.5), and N-antigen was detectable in 180

Table 1. Baseline Clinical Characteristics of Cohort Stratified by Disease Severity

Patient characteristics	Nonsevere	Severe	P value [*]
	N=43	N=45	
Demographics			
Age, y	57 (51–68)	61 (52–66)	0.491
Body mass index	30 (24–40)	31 (28–38)	0.430
Sex			0.526
Female	17 (39.5)	21 (46.7)	
Male	26 (60.5)	24 (53.3)	
Comorbidities			
Cardiac	25 (58.1)	27 (60.0)	>0.999
Diabetes	12 (27.9)	17 (37.8)	0.370
Chronic kidney disease	0 (0.0)	1 (2.2)	>0.999
Immunosuppression	4 (9.3)	2 (4.4)	0.429
Outcomes			
Hospital duration	5.0 (4.0, 9.0)	14.0 (9.0, 24.0)	<0.001
World Health Organization class			<0.001
4	14 (32.6)	0 (0.0)	
5	29 (67.4)	0 (0.0)	
6	0 (0.0)	38 (84.4)	
7	0 (0.0)	1 (2.2)	
8	0 (0.0)	3 (6.7)	
9	0 (0.0)	3 (6.7)	
90-d death	4 (9.3)	14 (31.1)	0.016
Thrombosis	3 (7.0)	10 (22.2)	0.070

Values are reported as median (interquartile range) or n (%). Biospecimens at the baseline time point were available for 88 of 93 total patients.

*Kruskal–Wallis rank-sum test; Fisher’s exact test.

(68%) samples, with a median level of 1687.5 pg/mL (interquartile range, 256–10071). RNA and N-antigen levels demonstrated moderate correlation ($r=0.51$, $P<0.001$; Figure S1).

RNA levels were higher at the time of trial enrollment in nonsurvivors compared with survivors regardless of disease severity classification (Figure 1A). In contrast, N-antigen levels were only significantly increased in nonsurvivors when all severities were included (Figure 1B). Temporally, RNA levels remained significantly higher in nonsurvivors compared with survivors through day 10 (Figure 1C, $P=7.75\times 10^{-5}$ for group interaction term Study Day*Mortality), whereas N-antigen levels were not significantly different (Figure 1D; $P=0.93$ for group interaction term Study Day*Mortality).

Plasma SARS-CoV-2 RNA Levels Are Associated With Thrombosis

Thirteen patients (14%) had clinically documented thrombotic events during their hospitalization. There was overlap between patients with documented

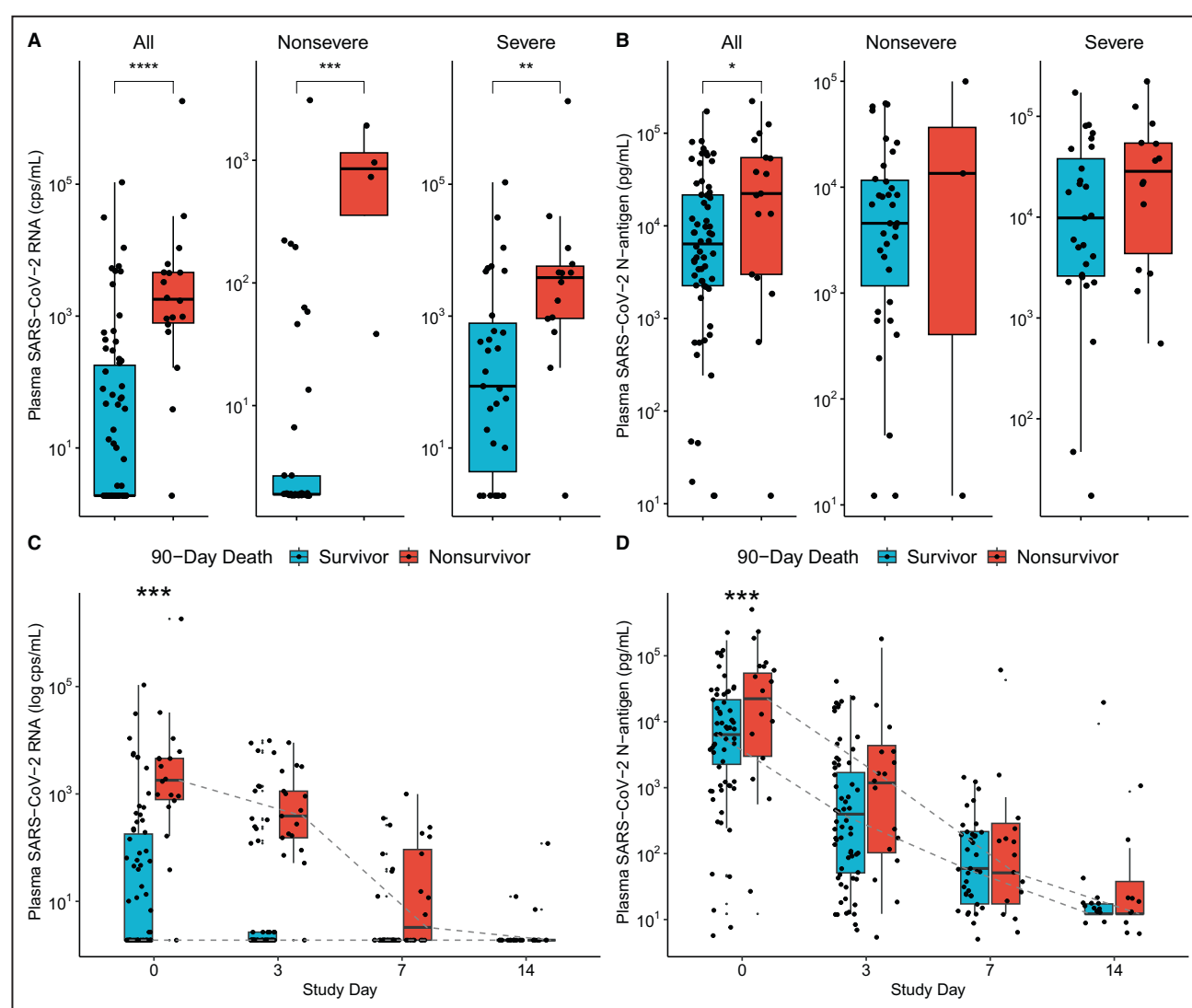


Figure 1. Plasma SARS-CoV-2 nucleocapsid RNA levels are higher in nonsurvivors across disease severity classification. **A, B,** Box and whisker plots showing plasma viral RNA (**A**) and N-antigen (**B**) levels, stratified by disease severity, and comparing 90-day death. Statistical comparisons were made using Wilcoxon rank-sum test. **C, D,** Box and whisker plots showing plasma viral RNA (**C**) or N-antigen (**D**) levels, stratified by sampling day and comparing nonsurvivors (blue) and survivors (orange). Dashed lines connect median values for each group (survivors vs nonsurvivors) at each sampling day to display longitudinal trends in viral RNA or N-antigen levels. In (**C**), sample size was distributed as follows: day 0: survivors (n=70), nonsurvivors (n=18); day 3: survivors (n=71), nonsurvivors (n=17); day 7: survivors (n=40), nonsurvivors (n=16); day 14: survivors (n=16), nonsurvivors (n=11). In (**D**), sample size was distributed as follows: day 0: survivors (n=62), nonsurvivors (n=17); day 3: survivors (n=63), nonsurvivors (n=17); day 7: survivors (n=37), nonsurvivors (n=15); day 14: survivors (n=14), nonsurvivors (n=10). Significance represented as follows: not significant (blank), * $P<0.05$, ** $P<0.01$, *** $P<0.001$.

thrombotic events and those who died (6/13 patients with thrombosis [46%]). RNA levels were higher in patients with documented thrombosis regardless of initial disease severity (Figure 2A). N-antigen levels were higher in patients with thrombosis only when comparing all patients in the cohort (Figure 2B). Temporally, RNA levels were higher throughout the study period in patients with thrombosis (Figure 2C; $P=1.24 \times 10^{-5}$ for group interaction term Study Day*Thrombosis). However, N-antigen temporal trends were not significantly different between patients with and without

thrombosis (Figure 2D; $P=0.72$ for group interaction term Study Day*Thrombosis).

Plasma SARS-CoV-2 RNA Levels Are Associated With Biomarkers of Thromboinflammation

Given the association between viral RNA and thrombotic events, we measured 9 key host-response biomarkers of thromboinflammation to determine their correlation with circulating SARS-CoV-2 RNA levels.

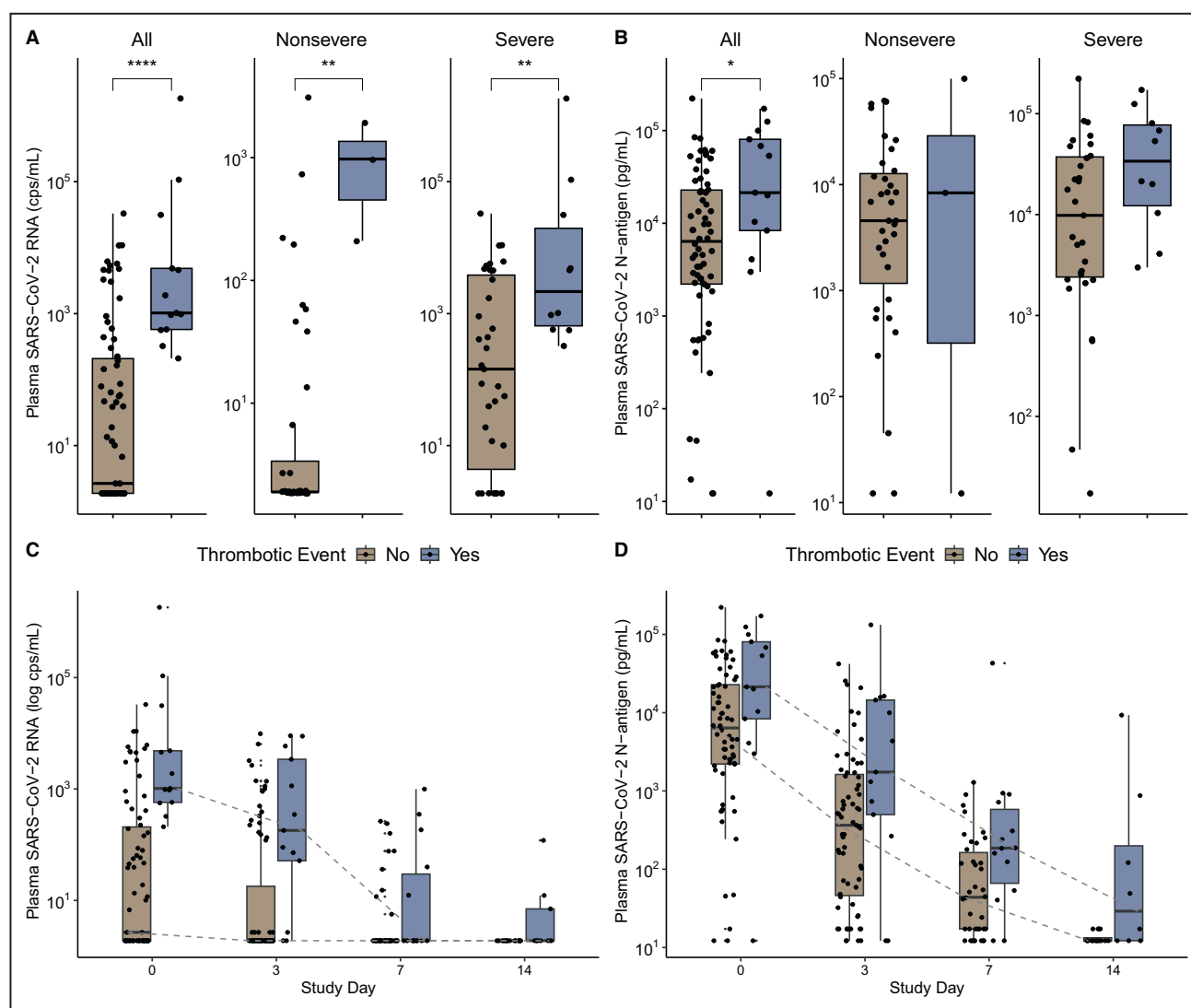


Figure 2. Plasma SARS-CoV-2 nucleocapsid RNA levels are higher in patients with thrombotic events across disease severity classifications.

A, B, Box and whisker plots showing plasma viral RNA (**A**) and N-antigen (**B**) levels, stratified by disease severity, and comparing incidence of thrombotic events. Statistical comparisons were made using Wilcoxon rank-sum test. **C, D,** Box and whisker plots showing plasma viral RNA (**C**) or N-antigen (**D**) levels, stratified by sampling day and comparing “yes” clot (purple) and “no” clot (brown). Dashed lines connect median values for each group (“yes” clot vs “no” clot) at each sampling day to display longitudinal trends in viral RNA or N-antigen levels. In (**C**), sample size was distributed as follows: day 0: “no” clot (n=75), “yes” clot (n=13); day 3: “no” clot (n=75), “yes” clot (n=13); day 7: “no” clot (n=42), “yes” clot (n=14); day 14: “no” clot (n=18), “yes” clot (n=9). In (**D**), sample size was distributed as follows: day 0: “no” clot (n=66), “yes” clot (n=13) as 9 samples were unavailable for N antigen; day 3: “no” clot (n=67), “yes” clot (n=13); day 7: “no” clot (n=38), “yes” clot (n=14); day 14: “no” clot (n=16), “yes” clot (n=8). Significance represented as follows: not significant (blank), * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

Distributions of all measured biomarkers across disease severities are included in Figure S2 and Table S3. As shown in Figure 3A, RNA levels significantly correlated with plasma levels of soluble receptor for advanced glycation end products (sRAGE), procalcitonin, pentraxin-3, interleukin-8, interleukin-6, interleukin-10, and fractalkine in the entire cohort. We found that RNA levels significantly correlated with sRAGE, pentraxin-3, and interleukin-6 in the nonsevere group. In the severe group, RNA levels correlated with plasma levels

of sRAGE and fractalkine. In contrast with thromboinflammation biomarkers, we found few significant correlations between plasma SARS-CoV-2 RNA levels and coagulation markers (Figure 3B). Notably, viral RNA levels only correlated with sTM, which is a membrane glycoprotein widely expressed on the luminal surface of endothelium³² that others have shown to be associated with worse clinical outcomes during COVID-19.^{24,33} Levels of tPA antigen, which is a serine protease expressed in endothelial cells that has been associated

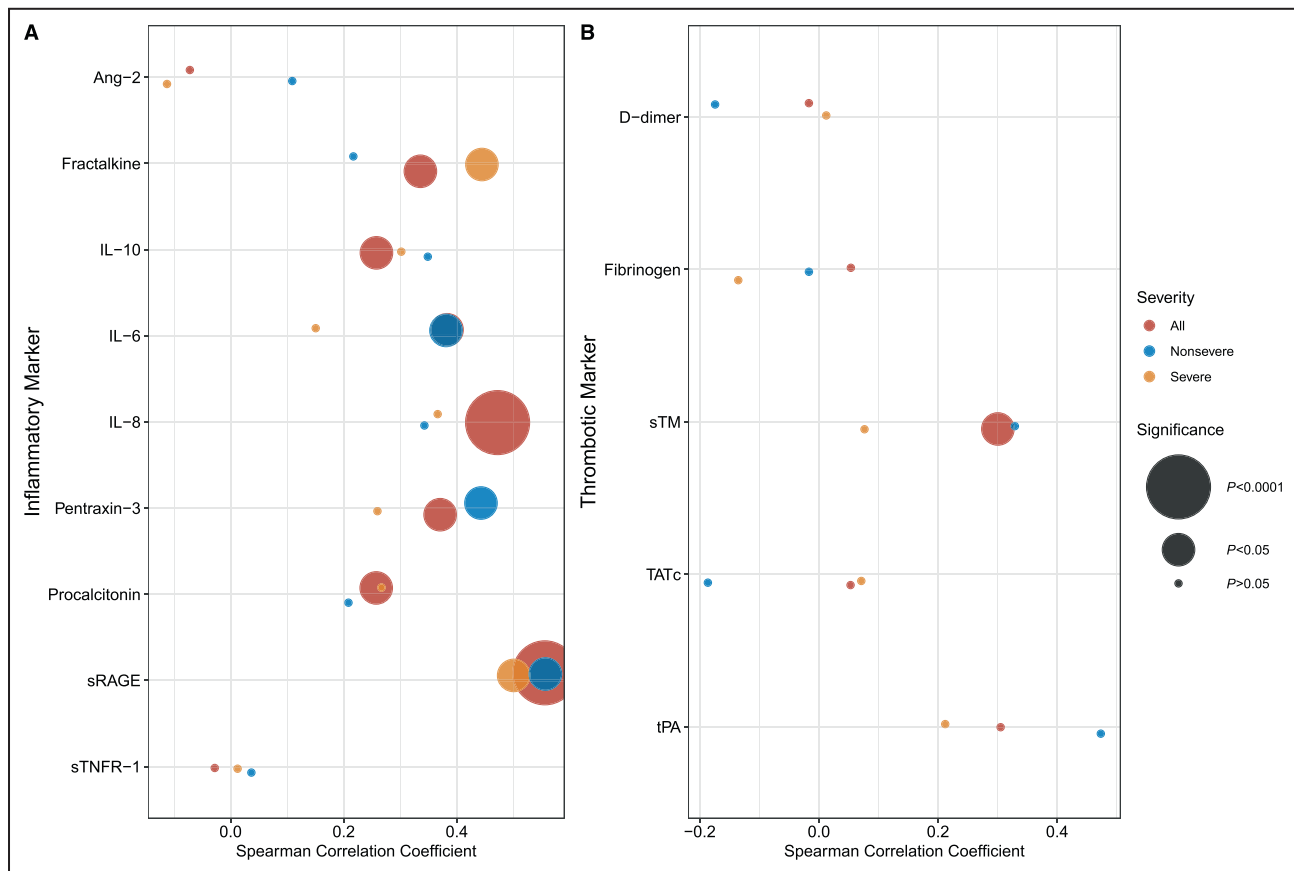


Figure 3. Plasma SARS-CoV-2 nucleocapsid RNA levels correlate with biomarkers of thromboinflammation.

A, B, Bubble plots showing correlation between cytokines (**A**) or thrombotic markers (**B**) and plasma viral RNA, stratified by disease severity. Points represent the Spearman correlation coefficient estimate, with point size indicating statistical significance. Note: P value=0.055 for the positive association between tPA and plasma RNA levels in All cohort after adjustment for multiple comparisons. Ang-2 indicates angiopoietin-2; IL, interleukin; sRAGE, soluble receptor for advanced glycation end products; sTM, soluble thrombomodulin; sTNFR, soluble tumor necrosis factor receptor; TATc, thrombin–antithrombin complexes; and tPA, tissue plasminogen activator.

with worse clinical outcomes during COVID-19,²⁵ correlated with plasma RNA levels (Spearman $r=0.31$, unadjusted $P=0.022$ in the entire cohort), although this association was no longer significant after adjustment for multiple comparisons ($P=0.056$), which may have resulted from fewer available samples ($n=56$). No other significant associations were noted. Based on this novel association between SARS-CoV-2 RNA levels and sTM, we sought to better understand the biological relationship between SARS-CoV-2 and sTM.

Thrombomodulin RNA transcripts are most numerous in lung and heart, and SARS-CoV-2 RNA transcripts are more likely in thrombomodulin-expressing cells during COVID-19 infection.

The gene that encodes thrombomodulin, *THBD*, is widely expressed in endothelial and other cells in tissues throughout the body.³² However, less is known about its expression during acute viral infection such as SARS-CoV-2. We used publicly available single-nucleus RNA-sequencing data performed on lung,

heart, liver, and kidney specimens collected during autopsy of 17 patients with fatal COVID-19.³⁴ Figure 4A demonstrates that *THBD* expression is more frequent in the lung (3.34% of all cells) and heart (2.64%) compared with liver (0.66%) and kidney (0.62%). Using assigned clusters and predicted cell types by automated annotation of lung specimens (Figure 4B; $N=99\,735$ total nuclei), *THBD* transcripts were most numerous in endothelial cells, both vascular and lymphatic, as well as mononuclear phagocytic cells, both macrophages and monocytes (Figure 4C). Cell-specific *THBD* transcripts were similarly enriched in endothelial and mononuclear phagocytic cells in the heart, liver, and kidney (Figure S3). Notably, viral transcripts, likely abortive replication, were reportedly more common in these cell types in the lung.³⁴ Interestingly, we found that cells with SARS-CoV-2 RNA transcripts are more likely to express *THBD* (Figure 4D). Taken together, thrombomodulin is transcribed in lung and heart endothelium and is associated with SARS-CoV-2 RNA, providing a

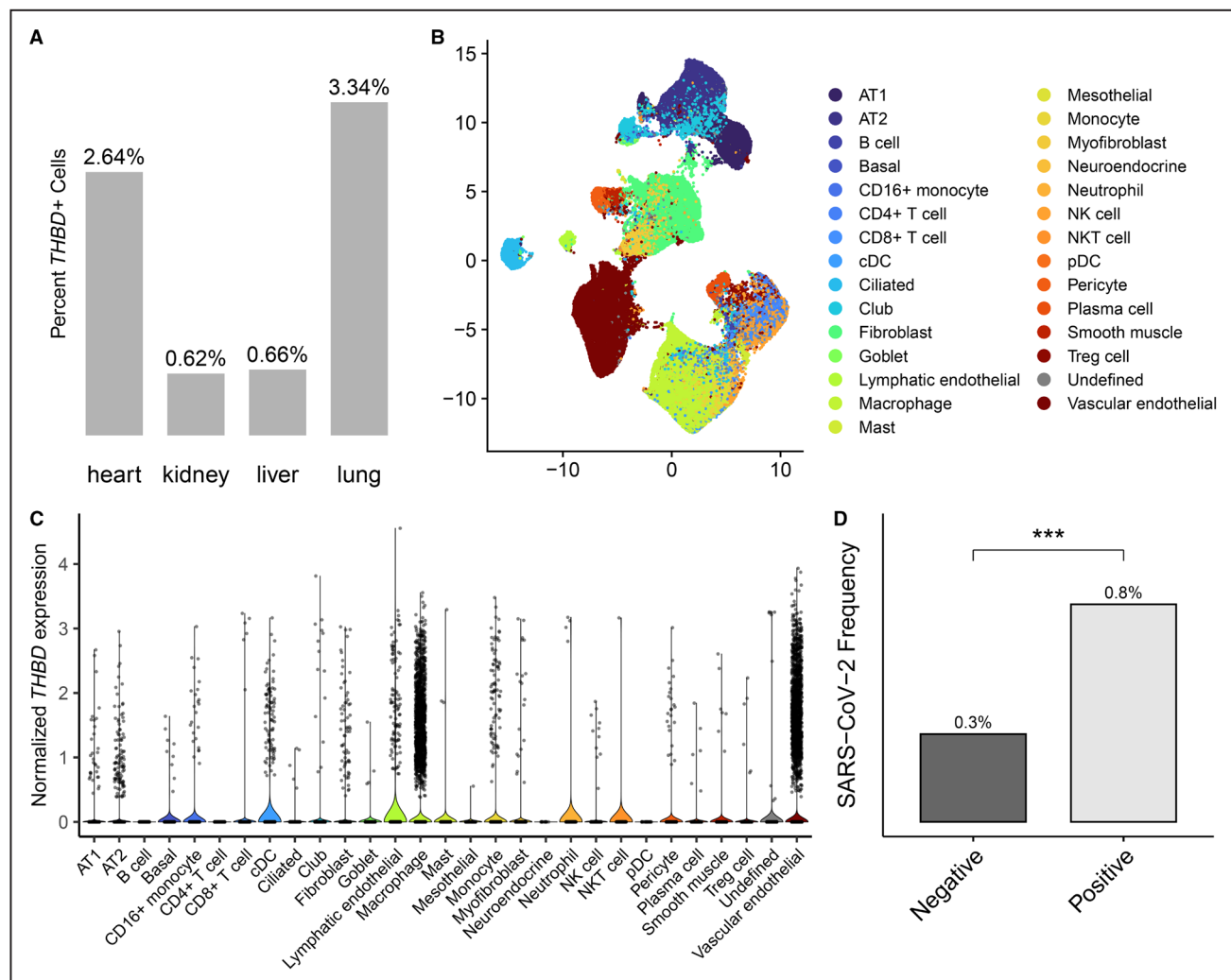


Figure 4. Thrombomodulin RNA transcripts are most numerous in the lung and heart and are more likely in lung cells with SARS-CoV-2 RNA during COVID-19.

A, Bar plot demonstrating expression of *THBD* gene transcript frequency by tissue type. **B**, Lung tissue Uniform Manifold Approximation and Projection with predicted cell types. **C**, Violin plot of *THBD* gene transcript levels by predicted cell type. **D**, Bar plot demonstrating SARS-CoV-2 RNA transcript frequency by *THBD* expression status. SARS-CoV-2 RNA transcripts were found in 27 of 3327 (0.81%) of *THBD*-expressing cells and 316 of 96408 (0.33%) of cells without *THBD* expression. In aggregate, 343 of 99735 (0.34%) of cells within the lung contained SARS-CoV-2 RNA transcripts. Significance represented as follows: not significant (blank), * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. AT1 indicates angiotensin II receptor type 1; AT2, angiotensin II receptor type 2; cDC, conventional dendritic cell; NK, natural killer; NKT, natural killer T; pDC, plasmacytoid dendritic cell; *THBD*, thrombomodulin gene; and Treg, T regulatory.

biologically plausible mechanism for its cardiovascular complications. Therefore, we sought to understand the relationship between viral RNA, sTM, and clinical outcomes.

Thrombosis, Plasma RNA Levels, and sTM Predict 90-Day Death

We used unadjusted and multivariable-adjusted time-to-event survival analysis to estimate the probability of 90-day death by thrombosis, plasma viral RNA and N-antigen levels, and coagulation markers. Adjusted estimates can be found in Table S4. In the unadjusted

models, clinically documented thrombosis was associated with an increased hazard ratio (HR) for 90-day death in all patients (HR, 3.44 [95% CI, 1.29–9.21]; $P = 0.014$) and patients in the nonsevere group (HR, 21.04 [95% CI, 2.86–154.49], $P = 0.003$), but not in patients in the severe group (HR, 1.41 [95% CI, 0.44–4.49]; $P = 0.57$) as shown in Table 2. Separately, for each 1-SD increase in SARS-CoV-2 plasma RNA levels, the HRs for 90-day death were 2.53 (95% CI, 1.7–3.77; $P < 0.001$) among all patients, 4.36 (95% CI, 1.57–12.08; $P = 0.005$) for patients in the nonsevere group, and 1.98 (95% CI, 1.18–3.33; $P = 0.01$) for patients in the severe group. Importantly, we did not

Table 2. Cox Proportional Hazards Ratios for Death Stratified by Disease Severity

Variable	All		Nonsevere		Severe	
	HR (95% CI)	P value	HR (95% CI)	P value	HR (95% CI)	P value
Thrombosis	3.44 (1.29–9.21)	0.014	21.04 (2.86–154.49)*	0.003	1.41 (0.44–4.49)	0.566
Viral RNA, log ₁₀ cps/mL	2.53 (1.7–3.77)	<0.001	4.36 (1.57–12.08)	0.005	1.98 (1.18–3.33)	0.010
N-antigen, log ₁₀ pg/mL	1.68 (0.9–3.13)	0.100	0.93 (0.31–2.83)	0.897	1.52 (0.79–2.92)	0.210
sTM, ng/mL	1.58 (1.15–2.16)	0.005	1.89 (1.12–3.18)	0.016	1.33 (0.86–2.06)	0.202
D-dimer, µg/mL	0.99 (0.6–1.64)	0.982	0.82 (0.21–3.26)	0.782	0.91 (0.48–1.75)	0.786
Fibrinogen, ng/mL	0.84 (0.52–1.36)	0.480	1.9 (0.76–4.71)	0.168	0.54 (0.31–0.97)	0.038
Thrombin–antithrombin complexes, ng/mL	0.8 (0.43–1.49)	0.484	0.3 (0–32.22)	0.614	0.95 (0.53–1.7)	0.853
tPA, ng/mL	1.25 (0.78–1.99)	0.356	1.39 (0.58–3.36)	0.459	1.13 (0.63–2.02)	0.675

All plasma biomarker levels correspond to the day 0 (baseline) time point. Baseline values were available for 88 of 93 patients. Variables reported on a logarithmic scale were log₁₀-transformed. Excluding thrombosis, all variables were standardized via Z score scaling before modeling. HRs reflect the risk associated with an increase of 1 SD for each scaled variable. Results displayed as estimated HR (95% CI). Sample sizes were distributed as follows: all: survivors (n=70), nonsurvivors (n=18), total (n=88); severe: survivors (n=31), nonsurvivors (n=14), total (n=45); nonsevere: survivors (n=39), nonsurvivors (n=4), total (n=43). For thrombosis, the sample sizes within each group were as follows: All: thrombosis (n=13), no thrombosis (n=75); severe: thrombosis (n=10), no thrombosis (n=35); nonsevere: thrombosis (n=3), no thrombosis (n=40). HR indicates hazard ratio; sTM, soluble thrombomodulin; and tPA, tissue plasminogen activator.

*Wide CI reflects low certainty due to small sample size. Absolute thrombosis incidence within the nonsevere group was as follows: survivors (n=1/39), nonsurvivors (3/4).

identify any statistically significant associations between N-antigen levels and 90-day death. We also identified statistically significant associations between increased sTM levels and death, as every 1-SD increase in sTM was associated with increased risk for 90-day death in all patients (HR, 1.58 [95% CI, 1.15–2.16]; $P=0.005$) and patients in the nonsevere group (HR, 1.89 [95% CI, 1.12–3.18]; $P=0.016$) but not in patients in the severe group (HR, 1.33 [95% CI, 0.86–2.06]; $P=0.2$). There was no significant interaction between sTM levels and treatment arm assignment on death. Notably, for every 1-SD increase in plasma fibrinogen, there was a decreased risk of 90-day death in patients in the severe group (HR, 0.54 [95% CI, 0.31–0.97]; $P=0.038$), which may reflect consumptive coagulopathy in some critically ill patients with COVID-19. Overall, these data indicate that both SARS-CoV-2 RNA and sTM levels

demonstrate a dose-dependent association with 90-day death. We next probed the association between thrombotic events and circulating viral RNA and coagulation markers.

Plasma RNA Levels and sTM Predict Thrombosis in Competing Risk Analysis

We used Fine–Gray competing risk analysis to estimate the probability of a thrombotic event while accounting for the competing risk of death. Adjusted analyses are available in Table S5. Unadjusted analyses shown in Table 3 demonstrate a robust cause-specific HR for thrombosis for each 1-SD increase in plasma RNA levels in all patients (HR, 2.84 [95% CI, 2.11–3.82]; $P<0.001$), patients in the nonsevere group (HR, 5.13 [95% CI, 2.14–12.31]; $P<0.001$), and in patients in the

Table 3. Fine–Gray Analysis of Thrombosis Accounting for Competing Risk of Death

Variable	All		Nonsevere		Severe	
	HR (95% CI)	P value	HR (95% CI)	P value	HR (95% CI)	P value
Viral RNA, log ₁₀ cps/mL	2.84 (2.11–3.82)	<0.001	5.13 (2.14–12.31)	<0.001	2.37 (1.69–3.34)	<0.001
N-antigen, log ₁₀ pg/mL	1.86 (0.66–5.24)	0.240	0.86 (0.19–3.85)	0.840	2.13 (0.96–4.71)	0.064
sTM, ng/mL	1.29 (0.93–1.81)	0.130	1.65 (1.09–2.52)	0.019	1 (0.58–1.71)	0.990
D-dimer, µg/mL	1.23 (0.75–2.02)	0.420	0.46 (0.05–4.32)	0.490	1.24 (0.67–2.3)	0.500
Fibrinogen, ng/mL	0.94 (0.52–1.7)	0.850	1.13 (0.22–5.86)	0.880	0.81 (0.46–1.42)	0.460
Thrombin–antithrombin complexes, ng/mL	0.48 (0.21–1.1)	0.082	0.01 (0–1.5)	0.069	0.52 (0.19–1.41)	0.200
tPA, ng/mL	1.2 (0.72–1.99)	0.480	2.27 (1.11–4.64)	0.024	0.94 (0.44–2.03)	0.880

All plasma biomarker levels correspond to the day 0 (baseline) time point. Baseline values were available for 88 of 93 patients. Variables reported on a logarithmic scale were log₁₀-transformed. Excluding thrombosis, all variables were standardized via Z score scaling before modeling. HRs reflect the risk associated with an increase of 1 SD for each scaled variable. Calculations performed using the “crr” Fine–Gray package in R. Results displayed as estimated HR (95% CI). HR indicates hazard ratio; sTM, soluble thrombomodulin; tPA, tissue plasminogen activator.

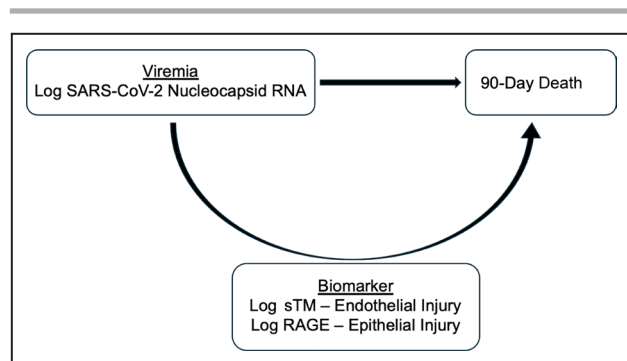


Figure 5. Directed acyclic graph depicting conceptual relationship between SARS-CoV-2 RNA and 90-day death (direct effect).

Curved arrow indicates possible mediation by endothelial dysfunction (measured as $\log_{10}$ levels of sTM) or epithelial damage (measured as $\log_{10}$ levels of soluble RAGE). RAGE indicates receptor for advanced glycation end products; and sTM, soluble thrombomodulin.

severe group (HR, 2.37 [95% CI, 1.69–3.34]; $P < 0.001$). We also found significant associations with thrombosis incidence in patients in the nonsevere group for every 1-SD increase in sTM (HR, 1.65 [95% CI, 1.09–2.52]; $P = 0.019$) and tPA (HR, 2.27 [95% CI, 1.11–4.64]; $P = 0.024$). As shown in Figure S4, survival for patients with thrombosis was worse in all patients ($P = 0.0087$) and in patients in the nonsevere group ($P < 0.0001$), but not in patients in the severe group ($P = 0.56$). Given the strong relationship between SARS-CoV-2 viremia, sTM levels, and death, we applied mediation models to assess for causal effects.

sTM Mediates a Portion of the Relationship Between Plasma Viremia and Death

We hypothesized that thrombotic endotheliopathy, as modeled by sTM, may mediate the relationship

between plasma viremia and death. Figure 5 shows a directed acyclic graph depicting this conceptual relationship. Based on our findings that sTM correlates with plasma RNAemia (Figure 3) and that both plasma RNA levels and sTM are associated with an increased risk of 90-day death (Table 2), we performed causal mediation analysis. We used sRAGE, a marker of lung epithelial injury that demonstrated the strongest correlation with plasma RNA levels (Figure 3A) as a biological control. In unadjusted analyses of the entire cohort, sTM significantly mediated a portion of the relationship between 90-day death and baseline viral RNA levels (proportion of mediated effect, 14.8% [95% CI, 0.9%–39.6%]), which remained significant (effect, 12.2% [95% CI, 0.3%–30.4%]) after adjustment for age and sex (Table 4). The trend was similar for sRAGE but did not reach statistical significance, as both unadjusted and adjusted models had CIs overlapping 0 (unadjusted 95% CI, –1.1% to 60.3%; adjusted 95% CI, –0.3% to 59.5%). Overall, these data suggest that sTM has a causal relationship that partially mediates the association between SARS-CoV-2 viremia and 90-day death.

DISCUSSION

We have identified a novel, potentially causal relationship between SARS-CoV-2 viremia, sTM levels, and death during hospitalization for COVID-19 in the ACTIV-4a randomized trial. We demonstrate that plasma SARS-CoV-2 RNA concentrations are significantly higher in nonsurvivors compared with survivors, whereas plasma SARS-CoV-2 N-antigen protein concentrations were not significantly different between survivors and nonsurvivors, despite being more sensitive for infection later in the disease course. We found a similar association between plasma RNA levels and

Table 4. Causal Mediation Analysis of Plasma SARS-CoV-2 RNA and Biomarkers Stratified by Disease Severity

Value	Log(sTM)				Log(RAGE)			
	Unadjusted		Adjusted		Unadjusted		Adjusted	
	Coefficient (95% CI)	P value	Coefficient (95% CI)	P value	Coefficient (95% CI)	P value	Coefficient (95% CI)	P value
Total effect	0.33 (0.178 to 0.489)	<2e-16	0.32 (0.196 to 0.473)	<2e-16	0.32 (0.162 to 0.457)	<2e-16	0.31 (0.155 to 0.452)	<2e-16
Average causal mediation effect	0.048 (0.003 to 0.118)	0.024	0.039 (0.001 to 0.099)	0.040	0.056 (–0.004 to 0.164)	0.058	0.05 (–0.01 to 0.141)	0.092
Average direct effect	0.28 (0.127 to 0.441)	0.002	0.28 (0.162 to 0.432)	<2e-16	0.26 (0.079 to 0.4)	0.002	0.26 (0.083 to 0.404)	0.002
Proportion mediated	0.15 (0.009 to 0.396)		0.12 (0.003 to 0.304)		0.18 (–0.011 to 0.603)		0.16 (–0.03 to 0.595)	

All results reported as estimate (95% CI). Adjusted model includes potential confounders of age and biological sex. Calculations performed using the R “mediate” package. RAGE indicates receptor for advanced glycation end products; and sTM, soluble thrombomodulin.

thrombosis, which is consistent with the overlap between patients with clinically documented thromboses and patients who died. Notably, patients with nonsevere SARS-CoV-2 infection had a significantly higher risk of death in the presence of a thrombotic event, while patients with severe infection did not. This is consistent with the findings of the REMAP-CAP (A Randomized, Embedded, Multifactorial, Adaptive Platform Trial for Community-Acquired Pneumonia), ACTIV-4a, and ATTACC (Antithrombotic Therapy to Ameliorate Complications of COVID-19) investigators, who found that therapeutic anticoagulation carries a mortality benefit in nonsevere SARS-CoV-2 infection but has no benefit in severe infection.^{21,22} We also found significant association of viral RNA with sTM, which is a marker of thrombotic endotheliopathy. We confirmed the biological plausibility of this association using publicly available transcriptomic data, which confirmed that thrombomodulin transcripts are most common in lung and heart cells, are frequently found in endothelial and monocytic cells, and are associated with viral RNA during infection. Finally, both SARS-CoV-2 viremia and sTM levels are associated with an increased risk for 90-day death and increased risk of thrombosis. Importantly, sTM mediates a portion of the relationship between viremia and death. Altogether, we interpret these findings to suggest that lung-blood translocation of intact SARS-CoV-2 virions may contribute to thrombotic endotheliopathy, providing a potential mechanism to understand the relationship between viral respiratory infection and its cardiovascular complications.

Our findings suggest that SARS-CoV-2 nucleocapsid RNA may better discriminate clinical outcomes than nucleocapsid protein levels despite their moderate correlation, which may point to important differences in their kinetics. Several studies have shown that nucleocapsid RNA and antigen associate with disease severity and poor outcome in acute COVID-19. The type and strength of association likely depends on timing, patient and disease characteristics, and differences between assay platforms.^{35,36} In studies that have examined both plasma viral RNA and antigen, the relationship between clinical outcomes and plasma antigen have been more complex than viral RNA. For example, in hospitalized patients, RNAemia was detectable earlier than antigenemia, and RNAemia at baseline was more strongly associated with nonsurvivorship than antigenemia, but any detectable antigenemia during hospitalization was also associated strongly with death.³⁵ In the ACTT-1 (Adaptive COVID-19 Treatment Trial), plasma viral RNA at baseline, but not plasma antigen at baseline, differentiated moderate/severe from critical disease as well as survivorship, but declines in RNAemia and antigenemia were both associated with decreased time to recovery and reduced death.³⁷ The complexity of detecting associations with plasma antigen levels

may reflect the dynamics of the SARS-CoV-2-specific immune response in the blood. Nucleocapsid-specific antibodies present in the blood could mask detection of nucleocapsid antigen to different degrees, depending on the relative levels of antibody to antigen, as was demonstrated in the context of antibody detection assays in blood.³⁸ Therefore, further work to understand the differences between detection of viral RNA and protein in the blood during COVID-19 may be beneficial for the use of viremia as a potential theragnostic marker. We further speculate that investigation of viremia during other respiratory viral infections such as influenza may help elucidate the connection between lower respiratory tract infections and their cardiovascular complications.

In addition to the discrepancy between viral RNA and protein levels, we note some complexity in markers of thrombotic endotheliopathy. In the current study, we observed a modest mediatory effect of plasma sTM on the relationship between plasma SARS-CoV-2 RNA levels and 90-day death at baseline. We also noted a correlation between tPA antigen levels and viral RNA levels, as well as thrombotic events, when accounting for the competing risk of death. Together, these findings suggest the presence of a thrombotic endotheliopathy contributing to both death and cardiovascular complications in COVID-19 infection. However, it is uncertain why levels of angiopoietin-2, another endothelial biomarker, were not associated with viral RNA levels in this cohort, despite the previously reported association of plasma angiopoietin-2 levels with intensive care unit admission in COVID-19.³⁹ Others have reported substantial variability in cytokine release during SARS-CoV-2 infection of lung microvascular endothelial cells from different donors in vitro, which may contribute to observed differences in biomarkers of endothelial injury.⁴⁰ Importantly, we note a strong correlation between plasma viral RNA levels and the marker of epithelial injury, sRAGE, although this association was not a significant mediator of the relationship between viremia and death. Therefore, the novel connection we have demonstrated between viremia and thrombotic endotheliopathy is worthy of further study, particularly for the potential impact on differential response to antithrombotic treatment, which we were underpowered to investigate in this study.

We note several limitations. First, we were provided biospecimens from only a subset of ACTIV-4 patients, and there is incomplete overlap in biospecimen availability for each specific assay reported herein. Therefore, our results may not reflect the entire trial population, and we may be underpowered to detect differences in some biomarkers. Additionally, thrombosis was identified by testing triggered by clinical decision making rather than protocolized screening, which may introduce bias in detection of thrombosis. Although we

did detect a mediatory effect of plasma sTM levels on death, a low event rate was observed, which may have impacted the significance of mediation analysis. We therefore used the penalized Firth method to mitigate the probability of false-positive analysis. We were provided clinical D-dimer levels obtained by the ACTIV-4a trial using heterogeneous methodologies, which may limit generalizability of our D-dimer findings.⁴¹ Finally, we have insufficient power to probe potential treatment-related effects of therapeutics such as anticoagulation therapy on viremia, thrombotic endotheliopathy, and their connection to death during COVID, which should be the focus of future investigations.

In conclusion, we present a novel, potentially causal association between SARS-CoV-2 viremia, sTM, and death in the ACTIV-4a randomized trial of antithrombotic therapy in patients hospitalized with COVID-19. This novel link identifies a potential mechanism for the relationship between lower respiratory tract viral infection and cardiovascular complications, which may guide future investigations.

ARTICLE INFORMATION

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Supplemental Material

Tables S1–S5

Figures S1–S4

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